Lab 4: VibeCoding - Chat History

Match-3 Gem Swap repair lab (57_match_three_gem) | 07 October 2026

Vibe coding tool: Claude Code, desktop app (model: Claude Opus 5.5)

This is the full conversation, in order: 6 prompts from the student, Claude's replies, and the 107 actions Claude took (commands run, files read and edited). Edits to the game's code are shown as diffs (+ added - removed). Long command outputs are trimmed, file contents that were only read are not repeated, and the tool's internal system messages are left out. Times are IST.

YOU - PROMPT 1 07 Oct 2026, 12:02

go through the pdf in this folder

Read `Lab_4_VibeCoding_Student_handout.pdf` **failed**

Listed files `**/*`

Read `Lab_4_VibeCoding_Student_handout.pdf`

Ran a command Check for Python and available PDF libraries

```
python --version 2>&1; python -c "import importlib.util as u; print({m: bool(u.find_spec(m)) for m in
['pypdf','PyPDF2','fitz','pdfplumber','pdfminer']})" 2>&1
```

```
Python 3.13.15
{'pypdf': False, 'PyPDF2': False, 'fitz': True, 'pdfplumber': False, 'pdfminer': False}
```

Read `README.md`

Read `main.py`

Read `game/board.py`

Read `game/game_engine.py`

Ran a command Show git remotes, recent commits, and installed game libraries

```
cd "C:/College/sem-v/se/lab4/57_match_three_gem" && git remote -v && git log --oneline -n 10 && python -c
"import importlib.util as u; print({m: bool(u.find_spec(m)) for m in
['pygame','tkinter','curses','numpy']})"
```

```
origin  https://github.com/SETAPESU26/57_match_three_gem.git (fetch)
origin  https://github.com/SETAPESU26/57_match_three_gem.git (push)
1126782 Update board.py
009c565 Update README.md
d5ccf05 Add files via upload
{'pygame': False, 'tkinter': True, 'curses': True, 'numpy': True}
```

Ran a command Show what the latest commit changed in board.py

```
cd "C:/College/sem-v/se/lab4/57_match_three_gem" && git show 1126782 --stat --format='%h %an %ad %s' && git
show 1126782 --format= -- game/board.py | head -60
```

```
1126782 Kritika-code Fri Oct 2 19:51:52 2026 +0530 Update board.py

 game/board.py | 16 +++-------------
 1 file changed, 3 insertions(+), 13 deletions(-)
diff --git a/game/board.py b/game/board.py
index 3527bdf..13237d1 100644
--- a/game/board.py
+++ b/game/board.py
@@ -19,7 +19,6 @@ class Gem:
```

```
         self.color = color
         self.target_row = target_row
         self.col = col
-        # Start higher up to animate falling down
         self.current_y = (target_row - 2) * TILE_SIZE
         self.target_y = target_row * TILE_SIZE
         self.fall_speed = 12.0
@@ -35,7 +34,6 @@ class Gem:


 class Board:
-    """Manages animated gem grid, gravity drops, score, and game limits."""

... (42 more lines not shown)
```

Ran a command Construct the board 300 times with pygame stubbed to see how often startup fails

```
cd "C:/College/sem-v/se/lab4/57_match_three_gem" && python -B -c "
import sys, types, collections
sys.modules['pygame'] = types.ModuleType('pygame')
from game.board import Board
res = collections.Counter()
for _ in range(300):
    try:
        Board(0, 0); res['ok'] += 1
... (4 more lines not shown)
```

```
{'ok': 26, "NameError: name 'c' is not defined": 274}
```

Ran a command Show the remainder of the latest board.py diff

```
cd "C:/College/sem-v/se/lab4/57_match_three_gem" && git show 1126782 --format= -- game/board.py | sed -n
'60,110p'
```

```
         for r in range(GRID_SIZE):
             for c in range(GRID_SIZE - 2):
                 if (
@@ -108,7 +101,6 @@ class Board:
                 ):
                     matched.update([(r, c), (r, c + 1), (r, c + 2)])

-        # Vertical matches
         for r in range(GRID_SIZE - 2):
             for c in range(GRID_SIZE):
                 if (
@@ -158,12 +150,10 @@ class Board:
         self.swap_gems(pos1, pos2)
         matches = self.find_matches()

-        # BUG SYMPTOM:
-        # Move count decrements on EVERY swap attempt even invalid ones.
         self.moves_remaining -= 1

         if not matches:
-            self.swap_gems(pos1, pos2)  # Revert invalid swap
+            self.swap_gems(pos1, pos2)
... (10 more lines not shown)
```

CLAUDE

I've finished reading the PDF, plus the README and the three source files it points to. I made no changes.

What the handout asks for

It's Lab 4 (VibeCoding), an individual 60-minute assignment: use an AI tool to fix the broken game and add the README's features.

1. Run the game and record a 10-second video **before any changes**.

2. Fix the broken code, then add each README feature one by one.
3. Record a second 10-second video after all the changes.
4. Push to a repo in your **personal** account, with a **separate commit per task**.
5. Do not raise a PR to the main repo (SETAPESU26).

Deliverables go in your Lab repo under a `Lab-4` folder: both videos, the updated code, and the chat history exported as a doc or PDF.

The tasks (from the README)

Task	What it is
1	Bug fix: an invalid swap reverts the gems but still costs a move
2	Cascade combo multiplier for chain reactions within one turn
3	4-in-a-row creates a glowing gem that clears a whole row or column when matched
4	Idle hint: after 5 seconds of no input, pulse an available swap pair

Things you should know before starting

- **The game crashes on startup most of the time.** `drop_and_refill` in `board.py:116` is missing its `for c in range(GRID_SIZE):` loop, so any match raises `NameError: name 'c' is not defined`. With pygame stubbed out, building the board failed 274 times out of 300. This is a second bug beyond the "one deliberate bug" the README mentions.
- **Even a launch that survives will crash on the first successful match**, for the same reason. An invalid swap won't crash, so the Task 1 bug can still be shown in the "before" video.
- **The Task 1 bug** is the unconditional `self.moves_remaining -= 1` at `board.py:153`, which runs before the match check.
- **pygame is not installed** on your Python 3.13, so the game can't run at all yet:

```
pip install pygame
```

- **`origin` points at `SETAPESU26/57_match_three_gem`**, the main repo. You'll need to point it at your own fork or personal repo before pushing.

Record the "before" video before I touch any code. Once that's done, tell me and I'll start with the crash fix and Task 1, one commit per task.

YOU - PROMPT 2 07 Oct 2026, 12:19

so I couldn't even run the game properly. it keeps crashing saying c is not defined in board.py under the game folder. is this a bug they have given and want us to solve?

Ran a command Check whether the column loop was missing in each commit of board.py

```
cd "C:/College/sem-v/se/lab4/57_match_three_gem" && for h in d5ccf05 009c565 1126782; do echo "== $h =="; git show $h:game/board.py | grep -n -A4 "def drop_and_refill"; done; git log --format='%h %an %ad %s' --date=short
```

```
== d5ccf05 ==
124:    def drop_and_refill(self):
125-            empty_slots = 0
126-            for r in range(GRID_SIZE - 1, -1, -1):
127-                if self.grid[r][c] is None:
128-                    empty_slots += 1
== 009c565 ==
124:    def drop_and_refill(self):
125-            empty_slots = 0
126-            for r in range(GRID_SIZE - 1, -1, -1):
127-                if self.grid[r][c] is None:
128-                    empty_slots += 1
== 1126782 ==
```

```
116:    def drop_and_refill(self):
117-            empty_slots = 0
118-            for r in range(GRID_SIZE - 1, -1, -1):
119-                if self.grid[r][c] is None:
120-                    empty_slots += 1
1126782 Kritika-code 2026-10-02 Update board.py
009c565 Kritika-code 2026-10-02 Update README.md
d5ccf05 Kritika-code 2026-09-25 Add files via upload
```

CLAUDE

Yes, treat it as theirs to fix: the crash is in the code exactly as the instructors uploaded it, not something you caused.

I checked the git history, and `drop_and_refill` is missing its column loop in all three commits, going back to the first upload on 2026-09-25. Whether it was intentional I can't tell from the repo. The README only claims "one deliberate bug" (the move deduction), and that one was originally marked with a `# BUG SYMPTOM` comment, while this one never had any marker. So it could be a second planted bug or an accident on their side.

Either way, fixing it is in scope: the handout says "fix the broken code", and none of the four tasks can be done or demonstrated while the game crashes.

The fix is one line at `board.py:116`. The method body is already indented one level too deep, which shows where the line was removed:

```
    def drop_and_refill(self):
        for c in range(GRID_SIZE):
            empty_slots = 0
            ...
```

Two things for your submission:

- **"Before" video:** record it before this fix goes in. The crash on launch is a legitimate "before", since it is the broken behaviour.
- **Commits:** give this fix its own commit, separate from Task 1, since the handout wants one commit per task.

Tell me once the video is recorded and I'll apply the fix.

YOU - PROMPT 3 07 Oct 2026, 12:23

I didn't fork the repository and then clone it. I just cloned the actual repository locally and I will committing to that. so after the fixes can we actually push these commits and fixed working code to my other software engineering repository for submission?

CLAUDE

Yes. Cloning directly is fine (the handout says "clone or fork"), and local commits don't care where they end up being pushed. Nothing is tied to the SETAPESU26 remote until you push, and we simply won't push there.

The plan I'd follow:

1. **Commit here, one commit per task**: crash fix, then Tasks 1–4.
2. **Bring this repo into your lab repo under** `Lab-4/` with `git subtree add`, run from inside the lab repo. That keeps every individual commit in the lab repo's history, which a plain copy-paste of the files would lose.
3. **Add the videos and the exported chat history** to `Lab-4/` and push the lab repo to your personal account.

The instructors' three original commits will come along in the history too, which is normal for a clone and shows what you started from.

Before any pushing, I'll also disable pushing on `origin` in this clone so nothing can go to the main repo by accident.

When we get to step 2, I'll need:

- the URL of your lab repo, or its local path if it's already cloned on this machine
- how your earlier labs are laid out in it (for example `Lab-3/` at the top level), so `Lab-4` matches

Neither blocks the fixes. Once your "before" video is recorded, say the word and I'll start with the crash fix.

YOU - PROMPT 4 07 Oct 2026, 18:12

i fixed the crash bug. I need to make a new commit for each of the task given in the readme.md file. help me do it all

Ran a command Check working tree state, recent commits, uncommitted diff, and whether pygame is installed

```
cd "C:/College/sem-v/se/lab4/57_match_three_gem" && git status --short --branch && echo "--- log ---" &&
git log --oneline -n 6 && echo "--- diff ---" && git diff && echo "--- libs ---" && python -c "import
importlib.util as u; print('pygame installed:', bool(u.find_spec('pygame')))"
```

```
## main...origin/main [ahead 1]
--- log ---
f109476 fixed the bug in board.py
1126782 Update board.py
009c565 Update README.md
d5ccf05 Add files via upload
--- diff ---
--- libs ---
pygame installed: True
```

Ran a command Inspect the user's crash-fix commit and list the folder

```
cd "C:/College/sem-v/se/lab4/57_match_three_gem" && git show f109476 --stat --format='%h %an %ad %s' && git
show f109476 --format= -- game/ main.py && ls -la && python -c "import pygame; print(pygame.version.ver)"
```

```
f109476 Arjun G Kanagal Wed Oct 7 18:10:31 2026 +0530 fixed the bug in board.py

 Lab_4_VibeCoding_Student_handout.pdf         | Bin 0 -> 71595 bytes
 game/__pycache__/board.cpython-313.pyc       | Bin 0 -> 11432 bytes
 game/__pycache__/game_engine.cpython-313.pyc | Bin 0 -> 5632 bytes
 game/board.py                                |   1 +
 4 files changed, 1 insertion(+)
diff --git a/game/__pycache__/board.cpython-313.pyc b/game/__pycache__/board.cpython-313.pyc
new file mode 100644
index 0000000..46fb6dd
Binary files /dev/null and b/game/__pycache__/board.cpython-313.pyc differ
diff --git a/game/__pycache__/game_engine.cpython-313.pyc b/game/__pycache__/game_engine.cpython-313.pyc
new file mode 100644
index 0000000..b2a1c98
Binary files /dev/null and b/game/__pycache__/game_engine.cpython-313.pyc differ
diff --git a/game/board.py b/game/board.py
index 13237d1..925b367 100644
--- a/game/board.py
+++ b/game/board.py
@@ -114,6 +114,7 @@ class Board:
         return matched

... (16 more lines not shown)
```

Read `game/board.py`

CLAUDE (PROGRESS NOTE)

Your crash fix is committed and pygame is installed, so I'll now work through Tasks 1–4, testing and committing each separately. First, I'll untrack the `__pycache__/*.pyc` files picked up by your commit and add a `.gitignore` so they don't pollute future commits.

Created file `.gitignore`

```
__pycache__/
*.pyc
```

Task 1: move deduction fix

Ran a command Untrack the pycache files, commit the gitignore, and confirm the files remain on disk

```
cd "C:/College/sem-v/se/lab4/57_match_three_gem" && git rm -r --cached --quiet game/__pycache__ && git add
.gitignore && git status --short && git commit --quiet -m "Stop tracking Python bytecode caches

Co-Authored-By: Claude Opus 5.5 <noreply@anthropic.com>" && git log --oneline -n 2 && ls game/__pycache__
```

```
warning: in the working copy of '.gitignore', LF will be replaced by CRLF the next time Git touches it
A  .gitignore
D  game/__pycache__/board.cpython-313.pyc
D  game/__pycache__/game_engine.cpython-313.pyc
826860b Stop tracking Python bytecode caches
f109476 fixed the bug in board.py
board.cpython-313.pyc
game_engine.cpython-313.pyc
```

CLAUDE

Now Task 1: move the deduction so it only happens after a swap is confirmed to make a match.

Edited `game/board.py`

```
@@ -1,5 +1,3 @@
         matches = self.find_matches()
-
-        self.moves_remaining -= 1

         if not matches:
@@ -7,3 +5,4 @@
             return False

+        self.moves_remaining -= 1
         cleared = self.resolve_matches()
```

Wrote a temporary helper script `scratch/harness.py` (74 lines, kept outside the repo)

Wrote a temporary helper script `scratch/test_task1.py` (35 lines, kept outside the repo)

Ran a command Run the Task 1 test with the fix, then against the unfixed code to confirm the test detects the bug

```
cd "C:/Users/LENOVO/AppData/Local/Temp/claude/C--College-sem-v-se-lab4-57-match-three-gem/85cc8ff0-3d18-
404c-9e6e-01e6b41e2cfc/scratchpad" && REPO="C:/College/sem-v/se/lab4/57_match_three_gem" && echo "== with
fix ==" && python -B test_task1.py; echo "== without fix (stashed) ==" && git -C "$REPO" stash --quiet &&
(python -B test_task1.py 2>&1 | tail -n 1); git -C "$REPO" stash pop --quiet && git
```

```
== with fix ==
task 1 OK: 200 invalid swaps free, 200 valid swaps cost 1 move
== without fix (stashed) ==
```

```
AssertionError: seed 0: invalid swap cost a move
 M game/board.py
```

Ran a command Commit the Task 1 fix

```
cd "C:/College/sem-v/se/lab4/57_match_three_gem" && git add game/board.py && git commit --quiet -m "Task 1: Deduct a move only when a swap creates a match

An invalid swap reverted the gems but still cost a move, because the
counter was decremented before the match check. Decrement it only after
the swap is confirmed to produce a match.

Co-Authored-By: Claude Opus 5.5 <noreply@anthropic.com>" && git log --oneline -n 3
```

```
1b203cc Task 1: Deduct a move only when a swap creates a match
826860b Stop tracking Python bytecode caches
f109476 fixed the bug in board.py
```

Task 2: cascade combo multiplier

CLAUDE (PROGRESS NOTE)

Task 1 done. Next up, I'll implement task 2: escalating chain-reaction multipliers (×1, ×2, ×3...) with a visible combo banner in the HUD.

Edited `game/board.py`

```
@@ -1,3 +1,5 @@
         self.moves_remaining = max_moves
+        self.last_points = 0
+        self.last_combo = 0
         self.reset()

@@ -5,3 +7,5 @@
         self.score = 0
         self.moves_remaining = self.max_moves
+        self.last_points = 0
+        self.last_combo = 0
         self.selected = None
```

Edited `game/board.py`

```
@@ -1,11 +1,13 @@
     def resolve_matches(self):
-        total_cleared = 0
+        points = 0
+        cascades = 0
         while True:
             matches = self.find_matches()
             if not matches:
                 break
-            total_cleared += len(matches)
+            cascades += 1
+            points += len(matches) * GEM_POINTS * cascades
             for r, c in matches:
                 self.grid[r][c] = None
             self.drop_and_refill()
-        return total_cleared
+        return points, cascades
```

Edited `game/board.py`

```
@@ -1,4 +1,4 @@
         self.moves_remaining -= 1
-        cleared = self.resolve_matches()
```

```
-         self.score += cleared * 10
+         self.last_points, self.last_combo = self.resolve_matches()
+         self.score += self.last_points
          return True
```

Edited `game/board.py`

```
@@ -1,3 +1,4 @@
 GRID_SIZE = 8
 TILE_SIZE = 60
+GEM_POINTS = 10
 GEM_COLORS = [
```

Edited `game/game_engine.py`

```
@@ -1,3 +1,4 @@
                 else:
-                    self.board.process_swap(prev_selected, (row, col))
+                    if self.board.process_swap(prev_selected, (row, col)):
+                        self.combo_time = pygame.time.get_ticks()
                     self.board.selected = None
```

Edited `game/game_engine.py`

```
@@ -1,2 +1,4 @@
         self.font_big = pygame.font.SysFont(None, 48)
         self.font_small = pygame.font.SysFont(None, 24)
+        self.font_combo = pygame.font.SysFont(None, 30)
+        self.combo_time = 0
```

Edited `game/game_engine.py`

```
@@ -1,3 +1,14 @@
         self.board.render(screen)

+        combo_age = pygame.time.get_ticks() - self.combo_time
+        if self.board.last_combo > 1 and combo_age < COMBO_BANNER_MS:
+            combo_surf = self.font_combo.render(
+                f"COMBO x{self.board.last_combo}!   +{self.board.last_points}",
+                True,
+                (255, 210, 70),
+            )
+            screen.blit(
+                combo_surf, (self.width // 2 - combo_surf.get_width() // 2, 84)
+            )
+
         inst_surf
```

Edited `game/game_engine.py`

```
@@ -1,2 +1,3 @@
 from game.board import Board, GRID_SIZE, TILE_SIZE

+COMBO_BANNER_MS = 2000
```

Wrote a temporary helper script `scratch/test_task2.py` (76 lines, kept outside the repo)

Wrote a temporary helper script `scratch/shot_task2.py` (30 lines, kept outside the repo)

Ran a command Run Task 1 and Task 2 tests and render a frame of the combo banner

```
cd "C:/Users/LENOVO/AppData/Local/Temp/claude/C--College-sem-v-se-lab4-57-match-three-gem/85cc8ff0-3d18-404c-9e6e-01e6b41e2cfc/scratchpad" && python -B test_task1.py && python -B test_task2.py && echo "--- screenshot ---" && python -B shot_task2.py | tail -n 2
```

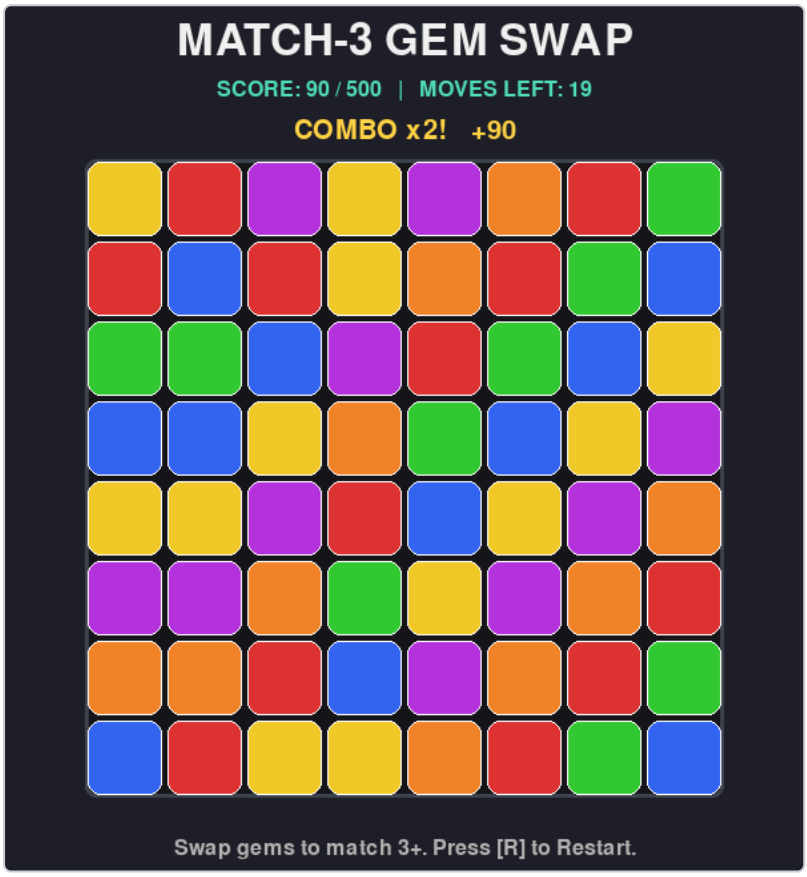
```
task 1 OK: 200 invalid swaps free, 200 valid swaps cost 1 move
forced cascade: sizes=[3, 3] -> 90 pts (flat would be 60), combo x2
```

```
task 2 OK: 190 single matches scored flat, 110 chain reactions scored with multiplier
--- screenshot ---
combo 2 points 90 moves 19
C:\Users\LENOVO\AppData\Local\Temp\claude\C--College-sem-v-se-lab4-57-match-three-gem\85cc8ff0-3d18-404c-9e6e-01e6b41e2cfc\scratchpad\task2.png
```

CLAUDE

Task 2 tests pass (a forced chain scored 90 instead of the old flat 60). Checking the rendered combo banner, then committing.

Read `scratch/task2.png` screenshot rendered by the test script



Ran a command Commit Task 2

```
cd "C:/College/sem-v/se/lab4/57_match_three_gem" && git status --short && git add game/board.py game/game_engine.py && git commit --quiet -m "Task 2: Add cascade combo multiplier

Each chain reaction within a turn now scores at a growing multiplier:
x1 for the player's own match, x2 for the first cascade, x3 for the
next, and so on. A single match scores exactly as before. The HUD shows
a short COMBO banner with the multiplier and points earned.

Co-Authored-By: Claude Opus 5.5 <noreply@anthropic.com>" && git log --oneline -n 3
```

```
 M game/board.py
 M game/game_engine.py
be00c1f Task 2: Add cascade combo multiplier
1b203cc Task 1: Deduct a move only when a swap creates a match
826860b Stop tracking Python bytecode caches
```

Task 3: line-clear special gems

> **Read** `game/board.py`

> **CLAUDE (PROGRESS NOTE)**
>
> Task 2 is committed. Next up is task 3: implementing the striped gem mechanic, where a 4+ match creates a gem that clears its row or column based on run direction, with chain blasts into other special gems.

> **Edited** `game/board.py`

```
@@ -1,2 +1,3 @@
+import math
 import random
 import pygame
@@ -5,2 +6,3 @@
 TILE_SIZE = 60
 GEM_POINTS = 10
+BLAST_FLASH_MS = 350
```

> **Edited** `game/board.py`

```
@@ -1,2 +1,3 @@
         self.target_y = target_row * TILE_SIZE
         self.fall_speed = 12.0
+        self.special = None
```

> **Edited** `game/board.py`

```
@@ -1,4 +1,5 @@
         self.last_points = 0
         self.last_combo = 0
+        self.blasts = []
         self.reset()

@@ -8,3 +9,4 @@
         self.last_points = 0
         self.last_combo = 0
+        self.blasts = []
         self.selected = None
```

> **Edited** `game/board.py`

```
@@ -1,5 +1,5 @@
                 self.grid[r][c] = gem

-        self.resolve_matches()
+        self.resolve_matches(spawn_specials=False)

     def is_animating(self):
```

> **Edited** `game/board.py`

```
@@ -1,24 +1,25 @@
+    def same_color(self, pos1, pos2):
+        g1 = self.grid[pos1[0]][pos1[1]]
+        g2 = self.grid[pos2[0]][pos2[1]]
+        return g1 is not None and g2 is not None and g1.color == g2.color
+
+    def find_runs(self):
+        lines = [("row", [(r, c) for c in range(GRID_SIZE)]) for r in range(GRID_SIZE)]
+        lines += [("col", [(r, c) for r in range(GRID_SIZE)]) for c in range(GRID_SIZE)]
+
+        runs = []
+        for orientation, line in lines:
```

```
+            start = 0
+            for i in range(1, GRID_SIZE + 1):
+                if i < GRID_SIZE and self.same_color(line[start], line[i]):
+                    continue
+                if i - start >= 3:
+                    runs.append((orientation, line[start:i]))
+                start = i
+        return runs
+
     def find_matches(self):
         matched = set()
-
-        for r in range(GRID_SIZE):
-            for c in range(GRID_SIZE - 2):
-                if (
-                    self.grid[r][c]
-                    and self.grid[r][c + 1]
-                    and self.grid[r][c + 2]
-                    and self.grid[r][c].color == self.grid[r][c + 1].color == self.grid[r][c + 2].color
-                ):
-                    matched.update([(r, c), (r, c + 1), (r, c + 2)])
-
-        for r in range(GRID_SIZE - 2):
-            for c in range(GRID_SIZE):
-                if (
-                    self.grid[r][c]
-                    and self.grid[r + 1][c]
-                    and self.grid[r + 2][c]
-                    and self.grid[r][c].color == self.grid[r + 1][c].color == self.grid[r + 2][c].color
-                ):
-                    matched.update([(r, c), (r + 1, c), (r + 2, c)])
-
+        for _, cells in self.find_runs():
+            matched.update(cells)
         return matched
```

Edited `game/board.py`

```
@@ -1,13 +1,44 @@
-    def resolve_matches(self):
+    def detonate_specials(self, cleared):
+        pending = [(r, c) for r, c in cleared if self.grid[r][c].special]
+        while pending:
+            r, c = pending.pop()
+            if self.grid[r][c].special == "row":
+                line = [(r, i) for i in range(GRID_SIZE)]
+                self.blasts.append(("row", r, pygame.time.get_ticks()))
+            else:
+                line = [(i, c) for i in range(GRID_SIZE)]
+                self.blasts.append(("col", c, pygame.time.get_ticks()))
+
+            for pos in line:
+                if pos not in cleared:
+                    cleared.add(pos)
+                    if self.grid[pos[0]][pos[1]].special:
+                        pending.append(pos)
+
+    def resolve_matches(self, swapped=(), spawn_specials=True):
         points = 0
         cascades = 0
         while True:
-            matches = self.find_matches()
-            if not matches:
+            runs = self.find_runs()
+            if not runs:
                 break
             cascades += 1
-            points += len(matches) * GEM_POINTS * cascades
-            for r, c in matches:
```

```
+
+            cleared = set()
+            for _, cells in runs:
+                cleared.update(cells)
+            self.detonate_specials(cleared)
+            points += len(cleared) * GEM_POINTS * cascades
+
+            if spawn_specials:
+                for orientation, cells in runs:
+                    if len(cells) >= 4:
+                        r, c = next((pos for pos in cells if pos in swapped), cells[1])
+                        self.grid[r][c].special = orientation
+                        cleared.discard((r, c))
+
+            for r, c in cleared:
                 self.grid[r][c] = None
             self.drop_and_refill()
+            swapped = ()
         return points, cascades
```

Edited `game/board.py`

```
@@ -1 +1 @@
-        self.last_points, self.last_combo = self.resolve_matches()
+        self.last_points, self.last_combo = self.resolve_matches(swapped=(pos1, pos2))
```

Edited `game/board.py`

```
@@ -2,3 +2,43 @@
                     self.grid[r][c].update()

+        now = pygame.time.get_ticks()
+        self.blasts = [b for b in self.blasts if now - b[2] < BLAST_FLASH_MS]
+
+    def render_special(self, surface, gem, x, y):
+        pulse = (math.sin(pygame.time.get_ticks() / 180) + 1) / 2
+        glow = pygame.Surface((TILE_SIZE, TILE_SIZE), pygame.SRCALPHA)
+        pygame.draw.rect(
+            glow,
+            (255, 255, 255, int(110 + 145 * pulse)),
+            glow.get_rect(),
+            width=5,
+            border_radius=12,
+        )
+        surface.blit(glow, (x, y))
+
+        mid = TILE_SIZE // 2
+        if gem.special == "row":
+            stripe = pygame.Rect(x + 9, y + mid - 4, TILE_SIZE - 18, 8)
+        else:
+            stripe = pygame.Rect(x + mid - 4, y + 9, 8, TILE_SIZE - 18)
+        pygame.draw.rect(surface, (255, 255, 255), stripe, border_radius=4)
+
+    def render_blasts(self, surface):
+        now = pygame.time.get_ticks()
+        for orientation, index, started in self.blasts:
+            fade = 1 - (now - started) / BLAST_FLASH_MS
+            if fade <= 0:
+                continue
+            if orientation == "row":
+                rect = pygame.Rect(
+                    self.offset_x, self.offset_y + index * TILE_SIZE, GRID_SIZE * TILE_SIZE, TILE_SIZE
+                )
+            else:
+                rect = pygame.Rect(
+                    self.offset_x + index * TILE_SIZE, self.offset_y, TILE_SIZE, GRID_SIZE * TILE_SIZE
+                )
+            flash = pygame.Surface(rect.size, pygame.SRCALPHA)
+            flash.fill((255, 255, 255, int(210 * fade)))
```

```
+            surface.blit(flash, rect.topleft)
+
     def render(self, surface):
```

Edited `game/board.py`

```
@@ -2,4 +2,6 @@
                         surface, (255, 255, 255), tile_rect, width=1, border_radius=10
                     )
+                    if gem.special:
+                        self.render_special(surface, gem, x, int(y))

                 if self.selected == (r, c):
@@ -10,2 +12,4 @@
                         surface, (255, 255, 255), sel_rect, width=4, border_radius=10
                     )
+
+        self.render_blasts(surface)
```

Wrote a temporary helper script `scratch/test_task3.py` (203 lines, kept outside the repo)

Wrote a temporary helper script `scratch/test_task2_layout.py` (10 lines, kept outside the repo)

Adjusted a temporary helper script `scratch/test_task3.py`

Adjusted a temporary helper script `scratch/test_task3.py`

Ran a command Run Task 1 regression and the Task 3 special-gem tests **failed**

```
cd "C:/Users/LENOVO/AppData/Local/Temp/claude/C--College-sem-v-se-lab4-57-match-three-gem/85cc8ff0-3d18-404c-9e6e-01e6b41e2cfc/scratchpad" && python -B test_task1.py && python -B test_task3.py
```

```
Exit code 1
task 1 OK: 200 invalid swaps free, 200 valid swaps cost 1 move
Traceback (most recent call last):
  File "C:\Users\LENOVO\AppData\Local\Temp\claude\C--College-sem-v-se-lab4-57-match-three-gem\85cc8ff0-3d18-404c-9e6e-01e6b41e2cfc\scratchpad\test_task3.py", line 5, in <module>
    import game.board as gb
ModuleNotFoundError: No module named 'game'
```

CLAUDE

Task 3 code is written; my test script just had an import-order slip (not a game bug). Fixing and re-running.

Adjusted a temporary helper script `scratch/test_task3.py`

Ran a command Run the Task 3 special-gem tests **failed**

```
cd "C:/Users/LENOVO/AppData/Local/Temp/claude/C--College-sem-v-se-lab4-57-match-three-gem/85cc8ff0-3d18-404c-9e6e-01e6b41e2cfc/scratchpad" && python -B test_task3.py
```

```
Exit code 1
find_matches matches the original scan on 2000 boards (incl. gaps, long runs)
300 fresh boards: no specials, no leftover matches
300 random games OK: at least 653 special gems created, 356 line blasts fired
horizontal 4-in-a-row -> row gem at the swapped cell, 40 pts
vertical 4-in-a-row -> column gem, kept through gravity, 40 pts
cascade-formed 4-run -> special gem on the run
Traceback (most recent call last):
  File "C:\Users\LENOVO\AppData\Local\Temp\claude\C--College-sem-v-se-lab4-57-match-three-gem\85cc8ff0-3d18-404c-9e6e-01e6b41e2cfc\scratchpad\test_task3.py", line 147, in <module>
    assert not (doomed & alive), "every gem in the blasted row must be gone"
           ^^^^^^^^^^^^^^^^^^^^
AssertionError: every gem in the blasted row must be gone
```

Ran a command Fix the test to compare gem objects instead of recycled ids, then re-run

```
cd "C:/Users/LENOVO/AppData/Local/Temp/claude/C--College-sem-v-se-lab4-57-match-three-gem/85cc8ff0-3d18-
404c-9e6e-01e6b41e2cfc/scratchpad" && python -B - <<'EOF'
import re, pathlib
p = pathlib.Path("test_task3.py")
s = p.read_text()
# Hold the gem objects themselves: id() values get recycled once a cleared gem is freed.
s = s.replace(
    "doomed = {id(board.grid[5][c]) for c in range(GRID_SIZE)} | {id(board.grid[4][2]), id(board.grid[6]
[3])}",
    "doomed = [board.grid[5][c] for c in range(GRID_SIZE)] + [board.grid[4][2], board.grid[6][3]]",
... (13 more lines not shown)
```

```
2 assertions rewritten
find_matches matches the original scan on 2000 boards (incl. gaps, long runs)
300 fresh boards: no specials, no leftover matches
300 random games OK: at least 653 special gems created, 356 line blasts fired
horizontal 4-in-a-row -> row gem at the swapped cell, 40 pts
vertical 4-in-a-row -> column gem, kept through gravity, 40 pts
cascade-formed 4-run -> special gem on the run
row gem matched -> entire row cleared (10 gems, 100 pts)
column gem matched -> entire column cleared (10 gems, 100 pts)
chained blast: row gem sets off a column gem (15 gems, 150 pts)
special gems only detonate when they are part of a match
cascade multiplier unchanged: 90 pts, combo x2
task 3 OK
```

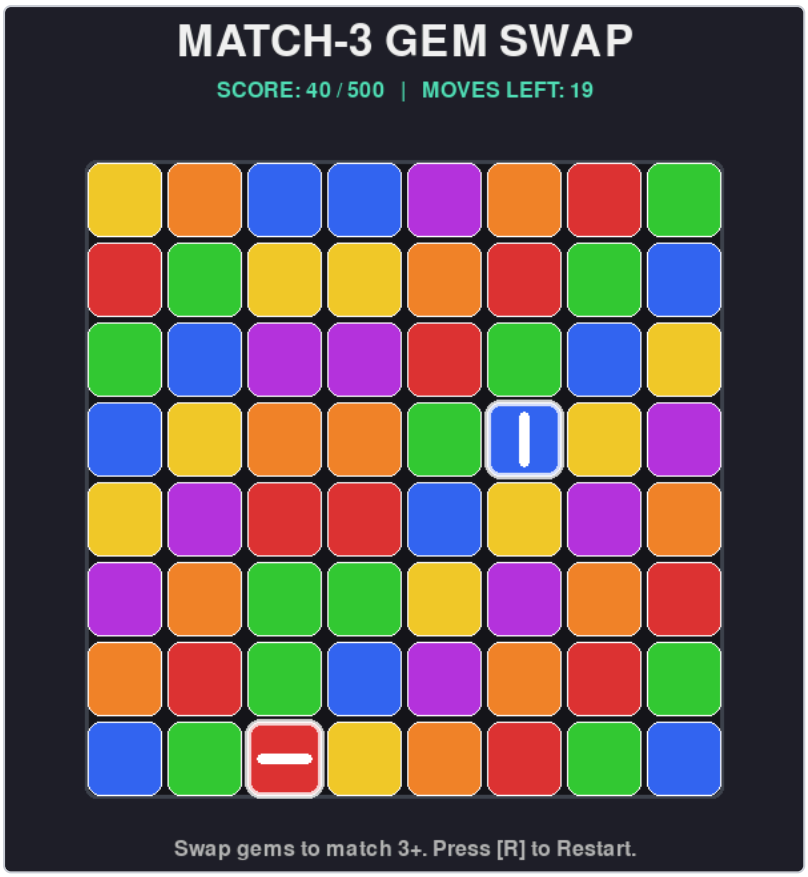
Wrote a temporary helper script `scratch/shot_task3.py` (48 lines, kept outside the repo)

Ran a command Render frames of the special gems and a line blast

```
cd "C:/Users/LENOVO/AppData/Local/Temp/claude/C--College-sem-v-se-lab4-57-match-three-gem/85cc8ff0-3d18-
404c-9e6e-01e6b41e2cfc/scratchpad" && python -B shot_task3.py
```

```
A: {(3, 5): 'col', (7, 2): 'row'} score 40
B: blasts [('row', 5)] score 100 moves 19
```

Read `scratch/task3_gems.png` screenshot rendered by the test script



Read `scratch/task3_blast.png` screenshot rendered by the test script



CLAUDE

Both visuals check out: striped glowing gems (frame 1) and the row flash when one is matched (frame 2). Committing Task 3.

Ran a command Commit Task 3

```
cd "C:/College/sem-v/se/lab4/57_match_three_gem" && git status --short && git add game/board.py && git
commit --quiet -m "Task 3: Add 4-in-a-row line-clear special gems

Matching 4 or more gems in a line now leaves a glowing striped gem
behind instead of clearing all of them. When that gem is later part of
a match it detonates its entire row (if made by a horizontal run) or
column (vertical run), and a blast that reaches another special gem
sets that one off too. Blasted gems score like matched gems and are
subject to the cascade multiplier.
... (6 more lines not shown)
```

```
 M game/board.py
2be7d50 Task 3: Add 4-in-a-row line-clear special gems
be00c1f Task 2: Add cascade combo multiplier
1b203cc Task 1: Deduct a move only when a swap creates a match
```

Task 4: idle hint indicator

Read `game/game_engine.py`

CLAUDE

Task 3 is committed. Task 4: after 5 seconds with no key press or mouse click, the board finds an available swap and softly pulses the two gems; any input clears it and restarts the timer.

Edited `game/board.py`

```
@@ -1,3 +1,4 @@
         self.blasts = []
+        self.hint = None
         self.reset()

@@ -8,3 +9,4 @@
         self.last_combo = 0
         self.blasts = []
+        self.hint = None
         self.selected = None
```

Edited `game/board.py`

```
@@ -1,3 +1,16 @@
         return matched

+    def find_hint(self):
+        grid = self.grid
+        for r in range(GRID_SIZE):
+            for c in range(GRID_SIZE):
+                for r2, c2 in ((r, c + 1), (r + 1, c)):
+                    if r2 < GRID_SIZE and c2 < GRID_SIZE:
+                        grid[r][c], grid[r2][c2] = grid[r2][c2], grid[r][c]
+                        makes_match = bool(self.find_runs())
+                        grid[r][c], grid[r2][c2] = grid[r2][c2], grid[r][c]
+                        if makes_match:
+                            return (r, c), (r2, c2)
+        return None
+
```

```
     def drop_and_refill(self):
```

Edited `game/board.py`

```
@@ -1,2 +1,3 @@
          self.moves_remaining -= 1
+         self.hint = None
          self.last_points, self.last_combo = self.resolve_matches(swapped=(pos1, pos2))
```

Edited `game/board.py`

```
@@ -1 +1,20 @@
+    def render_hint(self, surface):
+        pulse = (math.sin(pygame.time.get_ticks() / 300) + 1) / 2
+        for r, c in self.hint:
+            shimmer = pygame.Surface((TILE_SIZE, TILE_SIZE), pygame.SRCALPHA)
+            tile_rect = shimmer.get_rect().inflate(-4, -4)
+            pygame.draw.rect(
+                shimmer, (255, 255, 255, int(30 + 60 * pulse)), tile_rect, border_radius=10
+            )
+            pygame.draw.rect(
+                shimmer,
+                (255, 240, 150, int(80 + 175 * pulse)),
+                tile_rect,
+                width=3,
+                border_radius=10,
+            )
+            surface.blit(
+                shimmer, (self.offset_x + c * TILE_SIZE, self.offset_y + r * TILE_SIZE)
+            )
+
     def render_blasts(self, surface):
```

Edited `game/board.py`

```
@@ -1 +1,3 @@
+        if self.hint:
+            self.render_hint(surface)
          self.render_blasts(surface)
```

Edited `game/game_engine.py`

```
@@ -1,3 +1,5 @@
 COMBO_BANNER_MS = 2000
+HINT_DELAY_MS = 5000
+

 class GameEngine:
```

Edited `game/game_engine.py`

```
@@ -1,3 +1,10 @@
          self.combo_time = 0
+         self.last_input_time = pygame.time.get_ticks()
+         self.hint_searched = False
+
+    def register_input(self):
+         self.last_input_time = pygame.time.get_ticks()
+         self.hint_searched = False
+         self.board.hint = None

     def handle_click(self, mouse_pos):
```

Edited `game/game_engine.py`

```
@@ -1,2 +1,12 @@
     def update(self):
```

```
         self.board.update()
+
+        idle_time = pygame.time.get_ticks() - self.last_input_time
+        if (
+            not self.hint_searched
+            and idle_time > HINT_DELAY_MS
+            and not self.board.is_game_over()
+            and not self.board.is_animating()
+        ):
+            self.board.hint = self.board.find_hint()
+            self.hint_searched = True
```

Edited `main.py`

```
@@ -1,2 +1,5 @@
         for event in pygame.event.get():
+            if event.type in (pygame.KEYDOWN, pygame.MOUSEBUTTONDOWN):
+                engine.register_input()
+
             if event.type == pygame.QUIT:
```

Wrote a temporary helper script `scratch/test_task4.py` (182 lines, kept outside the repo)

Ran a command Run the Task 4 idle-hint tests **failed**

```
cd "C:/Users/LENOVO/AppData/Local/Temp/claude/C--College-sem-v-se-lab4-57-match-three-gem/85cc8ff0-3d18-404c-9e6e-01e6b41e2cfc/scratchpad" && python -B test_task4.py
```

```
Exit code 1
find_hint returned a valid adjacent swap on 300 boards without touching the grid
no available move -> no hint
Traceback (most recent call last):
  File "C:\Users\LENOVO\AppData\Local\Temp\claude\C--College-sem-v-se-lab4-57-match-three-gem\85cc8ff0-3d18-404c-9e6e-01e6b41e2cfc\scratchpad\test_task4.py", line 76, in <module>
    assert hint in valid, hint
           ^^^^^^^^^^^^^^
AssertionError: None
```

Adjusted a temporary helper script `scratch/test_task4.py`

Ran a command Re-run the Task 4 idle-hint tests

```
cd "C:/Users/LENOVO/AppData/Local/Temp/claude/C--College-sem-v-se-lab4-57-match-three-gem/85cc8ff0-3d18-404c-9e6e-01e6b41e2cfc/scratchpad" && python -B test_task4.py
```

```
find_hint returned a valid adjacent swap on 300 boards without touching the grid
no available move -> no hint
idle 5001ms -> hint ((0, 5), (0, 6)); nothing before that; searched once
input clears the hint; it returns after another 5s
clicking the hinted pair makes a match
hint waits for the board to settle
no hint on the game-over screen
restart clears the hint and the timer
board with no moves: searched once, no hint
main loop: 2 keys + 2 mouse presses registered, mouse movement ignored
task 4 OK
```

Wrote a temporary helper script `scratch/shot_task4.py` (29 lines, kept outside the repo)

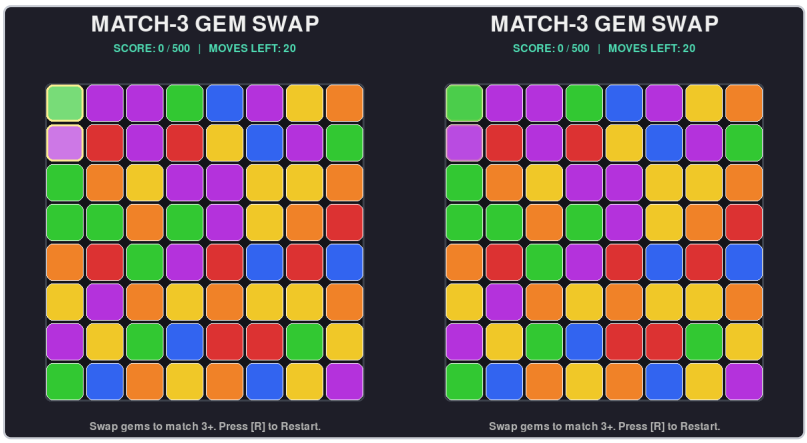
Ran a command Re-run Task 1 and Task 3 tests on the final code and render the hint

```
cd "C:/Users/LENOVO/AppData/Local/Temp/claude/C--College-sem-v-se-lab4-57-match-three-gem/85cc8ff0-3d18-404c-9e6e-01e6b41e2cfc/scratchpad" && python -B test_task1.py && python -B test_task3.py | tail -n 3 && python -B shot_task4.py
```

```
task 1 OK: 200 invalid swaps free, 200 valid swaps cost 1 move
special gems only detonate when they are part of a match
cascade multiplier unchanged: 90 pts, combo x2
```

```
task 3 OK
bright hint ((0, 0), (1, 0))
dim hint ((0, 0), (1, 0))
```

Read `scratch/task4.png` screenshot rendered by the test script



Ran a command Launch the real game window for 6 seconds as a smoke test

```
cd "C:/College/sem-v/se/lab4/57_match_three_gem" && timeout 6 python -B main.py; echo "exit code: $? (124 =
ran until the timeout stopped it)"; git status --short
```

```
exit code: 124 (124 = ran until the timeout stopped it)
 M game/board.py
 M game/game_engine.py
 M main.py
```

CLAUDE

Task 4 passes its tests and the real game window ran cleanly for 6 seconds. Committing it now.

Ran a command Commit Task 4 and show the final history

```
cd "C:/College/sem-v/se/lab4/57_match_three_gem" && git add game/board.py game/game_engine.py main.py &&
git commit --quiet -m "Task 4: Add idle hint indicator

After more than 5 seconds without a key press or mouse click, the board
looks for an adjacent swap that would make a match and highlights the
two gems with a soft pulsing shimmer. Any input hides the hint and
restarts the timer. No hint is shown while gems are falling or once the
game is over.

... (1 more lines not shown)
```

```
743cbc9 Task 4: Add idle hint indicator
2be7d50 Task 3: Add 4-in-a-row line-clear special gems
be00c1f Task 2: Add cascade combo multiplier
1b203cc Task 1: Deduct a move only when a swap creates a match
826860b Stop tracking Python bytecode caches
f109476 fixed the bug in board.py
1126782 Update board.py
009c565 Update README.md
--- status ---
## main...origin/main [ahead 6]
```

CLAUDE

All four tasks are done, each in its own commit on `main`. Nothing has been pushed.

Commit	What it does
`826860b`	Housekeeping: stops tracking `__pycache__` and adds a `.gitignore`
`1b203cc`	**Task 1**: a move is deducted only when the swap makes a match
`be00c1f`	**Task 2**: cascade multiplier (×1 for your match, ×2 for the first chain reaction, ×3 for the next), with a yellow `COMBO x2! +90` banner under the score
`2be7d50`	**Task 3**: 4+ in a line leaves a glowing striped gem; matching it later clears its whole row or column
`743cbc9`	**Task 4**: after 5 seconds with no click or key press, an available swap pair pulses softly

Your own crash fix (`f109476`) sits before these, untouched.

How it behaves

- **Special gems:** a horizontal run of 4 makes a gem with a horizontal stripe that clears its row; a vertical run makes a vertical-stripe gem that clears its column. A blast that hits another special gem sets that one off too, and the cleared line flashes white briefly.
- **Hint:** mouse movement alone doesn't count as input, so the hint still appears while you're hovering and looking. It never shows while gems are falling or on the win/loss screen.

Verification

I tested each task with headless scripts before committing, rendered frames to check the banner, special gems, blast flash and hint by eye, and ran the real game window for 6 seconds without errors. I have not played it by hand, so give it a quick play before recording.

Things I left alone

- If a board ever has no possible move, the game doesn't reshuffle; no hint appears and you have to press R. That was already the case.
- Falling gems briefly draw over the title text at the top. This is also original behaviour.

What's left for the submission

1. Record the "after" video. Ten seconds is tight, so the 5-second idle wait for the hint will take half of it.
2. Export this chat as a doc or PDF.
3. Push to your lab repo under `Lab-4/`. For that I need the lab repo's URL (or local path) and how your earlier labs are laid out in it.

YOU - PROMPT 5 07 Oct 2026, 19:01

this is the path to the actual github repository I have cloned locally: C:\College\sem-v\se\github\SoftwareEngineering\lab4. this is lab4 (a separate folder in that repo) where we have to push this changed code along with the videos and our chat as a pdf.

Ran a command Inspect the lab repository: remotes, branch, status, layout, and the lab4 folder contents

```
cd "C:/College/sem-v/se/github/SoftwareEngineering" && echo "--- toplevel ---" && git rev-parse --show-toplevel && echo "--- remotes ---" && git remote -v && echo "--- branch/status ---" && git status --short --branch && echo "--- log ---" && git log --oneline -n 8 && echo "--- top level ---" && ls -la && echo "--- lab4 ---" && ls -la lab4 2>&1 && echo "--- tracked under lab4 ---" && git ls-files
```

```
--- toplevel ---
C:/College/sem-v/se/github/SoftwareEngineering
--- remotes ---
origin  https://github.com/ArjunGK28/SoftwareEngineering.git (fetch)
origin  https://github.com/ArjunGK28/SoftwareEngineering.git (push)
--- branch/status ---
## main...origin/main
--- log ---
4f95a95 Add files via upload
```

```
87d4282 lab2 pdf
4f6a6f9 flow and table in pdf format
1295a27 UML diagram
666d4da add requirements table and diagram
--- top level ---
total 16
drwxr-xr-x 1 LENOVO 197609 0 Oct  7 18:39 .
drwxr-xr-x 1 LENOVO 197609 0 Sep 15 14:30 ..
drwxr-xr-x 1 LENOVO 197609 0 Oct  7 19:02 .git
drwxr-xr-x 1 LENOVO 197609 0 Aug 30 22:27 lab1
drwxr-xr-x 1 LENOVO 197609 0 Aug 30 22:28 lab2
drwxr-xr-x 1 LENOVO 197609 0 Oct  7 18:04 lab3
drwxr-xr-x 1 LENOVO 197609 0 Oct  7 18:39 lab4
... (6 more lines not shown)
```

Ran a command Fetch the lab repo's remote, check it is in sync, look at earlier lab layouts, and confirm git subtree is available

```
cd "C:/College/sem-v/se/github/SoftwareEngineering" && git fetch --quiet origin && echo "--- after fetch --
-" && git status --short --branch && git rev-list --left-right --count main...origin/main && echo "---
earlier labs ---" && for d in lab1 lab2 lab3; do echo "[$d]"; ls "$d"; done && echo "--- subtree available?
---" && (git subtree 2>&1 | head -n 2) && echo "--- autocrlf ---" && git config --
```

```
--- after fetch ---
## main...origin/main
0       0
--- earlier labs ---
[lab1]
Requirements_Table.docx
UseCase_Diagram_arjun.pdf
UseCase_Flow_UC06.docx
UseCase_Flow_UC06_arjun.pdf
requirements_table_arjun.pdf
[lab2]
report_PES1UG24CS922.pdf
[lab3]
Component_Lab3.pdf
Justificationdoc.pdf
--- subtree available? ---
usage: git subtree add   --prefix=<prefix> [-S[=<key-id>]] <commit>
   or: git subtree add   --prefix=<prefix> [-S[=<key-id>]] <repository> <ref>
--- autocrlf ---
true
--- lab folder next to the game clone ---
total 16848
... (7 more lines not shown)
```

Ran a command Compare commit times with the video timestamps and look at the 57_before folder

```
cd "C:/College/sem-v/se/lab4/57_match_three_gem" && echo "--- commit times ---" && git log --format='%h
%ad  %s' --date=format:'%Y-%m-%d %H:%M' -n 7 && echo "--- 57_before ---" && ls -la ../57_before && echo "--
- is 57_before a git repo? ---" && (git -C ../57_before rev-parse --show-toplevel 2>&1; git -C ../57_before
log --oneline -n 3 2>&1) && echo "--- lab repo identity ---" && git -C "C:/Colle
```

```
--- commit times ---
743cbc9  2026-10-07 18:23  Task 4: Add idle hint indicator
2be7d50  2026-10-07 18:21  Task 3: Add 4-in-a-row line-clear special gems
be00c1f  2026-10-07 18:16  Task 2: Add cascade combo multiplier
1b203cc  2026-10-07 18:15  Task 1: Deduct a move only when a swap creates a match
826860b  2026-10-07 18:14  Stop tracking Python bytecode caches
f109476  2026-10-07 18:10  fixed the bug in board.py
1126782  2026-10-02 19:51  Update board.py
--- 57_before ---
total 89
drwxr-xr-x 1 LENOVO 197609     0 Oct  7 18:29 .
drwxr-xr-x 1 LENOVO 197609     0 Oct  7 18:53 ..
-rw-r--r-- 1 LENOVO 197609    77 Oct  7 18:29 .git
-rw-r--r-- 1 LENOVO 197609 71595 Oct  7 18:29 Lab_4_VibeCoding_Student_handout.pdf
```

```
-rw-r--r-- 1 LENOVO 197609  3603 Oct  7 18:29 README.md
drwxr-xr-x 1 LENOVO 197609     0 Oct  7 18:29 game
-rw-r--r-- 1 LENOVO 197609   954 Oct  7 18:29 main.py
--- is 57_before a git repo? ---
C:/College/sem-v/se/lab4/57_before
f109476 fixed the bug in board.py
1126782 Update board.py
009c565 Update README.md
... (3 more lines not shown)
```

Import into lab repo

Ran a command Disable pushing to the course repo from the game clone, then import the game repo into the lab repo under lab4 and verify the contents match

```
cd "C:/College/sem-v/se/lab4/57_match_three_gem" && git remote set-url --push origin DISABLED && echo "--- game clone remotes ---" && git remote -v && cd "C:/College/sem-v/se/github/SoftwareEngineering" && rmdir lab4 && git subtree add --prefix=lab4 -m "Add Lab 4 match-three gem game under lab4/ with its commit history

Imported from the local clone of SETAPESU26/57_match_three_gem at
743cbc9, keeping the separate commits for the crash fix and Tasks 1-4.

Co-Authored-By: Claude Opus 5.5 <noreply@anthropic.com>" "C:/College/sem-v/se/lab4/57_match_three_gem" main 2>&1 | tail -n 5; echo "--- status ---" && git status --short --branch && echo "--- tree check ---" && echo "lab4 tree in lab repo : $(git rev-parse HEAD:lab4)" && echo "game repo tree at tip: $(git -C "C:/College/sem-v/se/lab4/57_match_three_gem" rev-parse 'main^{tree}')" && echo "--- lab4
```

```
--- game clone remotes ---
origin  https://github.com/SETAPESU26/57_match_three_gem.git (fetch)
origin  DISABLED (push)
git fetch C:/College/sem-v/se/lab4/57_match_three_gem main
From C:/College/sem-v/se/lab4/57_match_three_gem
 * branch            main       -> FETCH_HEAD
Added dir 'lab4'
--- status ---
## main...origin/main [ahead 10]
--- tree check ---
lab4 tree in lab repo : 6825c56da6445b6ec9c6d8613bf3633956e815f3
game repo tree at tip: 6825c56da6445b6ec9c6d8613bf3633956e815f3
--- lab4 contents ---
lab4:
total 89
drwxr-xr-x 1 LENOVO 197609     0 Oct  7 19:03 .
drwxr-xr-x 1 LENOVO 197609     0 Oct  7 19:03 ..
-rw-r--r-- 1 LENOVO 197609    21 Oct  7 19:03 .gitignore
-rw-r--r-- 1 LENOVO 197609 71595 Oct  7 19:03 Lab_4_VibeCoding_Student_handout.pdf
-rw-r--r-- 1 LENOVO 197609  3603 Oct  7 19:03 README.md
drwxr-xr-x 1 LENOVO 197609     0 Oct  7 19:03 game
-rw-r--r-- 1 LENOVO 197609  1069 Oct  7 19:03 main.py
... (8 more lines not shown)
```

Ran a command Copy the two videos into lab4, commit them, and show the resulting lab repo history

```
cd "C:/College/sem-v/se/github/SoftwareEngineering" && cp "C:/College/sem-v/se/lab4/initial_recording_cnotdefined.mp4" "C:/College/sem-v/se/lab4/after_fixing_recording.mp4" lab4/ && git add lab4/initial_recording_cnotdefined.mp4 lab4/after_fixing_recording.mp4 && git status --short && git commit --quiet -m "Add Lab 4 before and after gameplay videos

initial_recording_cnotdefined.mp4 shows the original crash on launch;
after_fixing_recording.mp4 shows the game after the fixes and features.

Co-Authored-By: Claude Opus 5.5 <noreply@anthropic.com>" && echo "--- history (lab repo) ---" && git log --oneline --graph -n 13 && echo "--- status ---" && git status --short --branch && echo "--- lab4 files tracked ---" && git ls-files -s lab4 | awk '{print $4}' && cmp "C:/College/sem-
```

```
v/se/lab4/after_fixing_recording.mp4" lab4/after_fixing_recording.mp4 && cmp "C:/College/sem-
v/se/lab4/initial_r
```

```
A  lab4/after_fixing_recording.mp4
A  lab4/initial_recording_cnotdefined.mp4
--- history (lab repo) ---
* 10b9d66 Add Lab 4 before and after gameplay videos
*   36ee2f4 Add Lab 4 match-three gem game under lab4/ with its commit history
|\
| * 743cbc9 Task 4: Add idle hint indicator
| * 2be7d50 Task 3: Add 4-in-a-row line-clear special gems
| * be00c1f Task 2: Add cascade combo multiplier
| * 1b203cc Task 1: Deduct a move only when a swap creates a match
| * 826860b Stop tracking Python bytecode caches
| * f109476 fixed the bug in board.py
| * 1126782 Update board.py
| * 009c565 Update README.md
| * d5ccf05 Add files via upload
* 4f95a95 Add files via upload
* 87d4282 lab2 pdf
--- status ---
## main...origin/main [ahead 11]
--- lab4 files tracked ---
lab4/.gitignore
lab4/Lab_4_VibeCoding_Student_handout.pdf
... (8 more lines not shown)
```

CLAUDE

The code and both videos are now committed locally in your lab repo under `lab4/`; nothing is pushed yet. Pushing publishes to your GitHub, so I need your go-ahead for that step.

Claude asked a question

Everything is committed locally in SoftwareEngineering/lab4 (game code with all task commits, plus both videos). How do you want to handle the push and the chat PDF?

- **Push now, chat PDF after (Recommended)** - I push the code and videos to ArjunGK28/SoftwareEngineering now. Then I export this chat, convert it to a PDF, add it to lab4/ as one more commit and push that too.
- **Push now, I'll do the PDF** - I push the code and videos now. You export the chat yourself and add the PDF to lab4/ later.
- **Don't push yet** - Leave everything committed locally so you can look it over first. Nothing goes to GitHub.

YOU (ANSWER)

I will give you the videos here. push it along with other things

CLAUDE

Understood: one push with everything together, so I haven't pushed anything.

One thing to check first. I had already found two videos sitting next to the game folder in `C:\College\sem-v\se\lab4\` and committed copies of them into `lab4/` locally:

- `initial_recording_cnotdefined.mp4` (14.7 MB, recorded 12:15)
- `after_fixing_recording.mp4` (2.5 MB, recorded 18:53, which is after the last task commit)

Are those the videos you meant? If so, there is nothing more to give me. If you have different ones, tell me their path or drop them in that same folder, and I'll swap them in before pushing; the commit is local, so replacing them is clean.

Where things stand in your lab repo (all local)

- `lab4/` holds the game code, with the crash fix and Tasks 1–4 as separate commits in the history.
- The `lab4/` contents are byte-identical to the game repo.
- The two videos above are one further commit on top.
- Your game clone can no longer push to the SETAPESU26 repo by accident (undo with `git config --unset remote.origin.pushurl` if you ever need to).

What's left

1. You confirm or replace the videos.
2. I export this chat, convert it to a PDF and add it to `lab4/`. I haven't tried the conversion yet.
3. I push everything to `ArjunGK28/SoftwareEngineering` in one go.

YOU - PROMPT 6 07 Oct 2026, 19:44

> Are those the videos you meant? If so, there is nothing more to give me. If you have different ones, tell me their path or drop them in that same folder, and I'll swap them in before pushing; the commit is local, so replacing them is clean.

yes

Exported this chat to build this PDF

Chat PDF and push

Ran a command Unpack the exported transcript into the scratchpad and list its files

```
SP="C:/Users/LENOVO/AppData/Local/Temp/claude/C--College-sem-v-se-lab4-57-match-three-gem/e3dd8c12-86c1-477c-96a4-36c9acf65a74/scratchpad" && mkdir -p "$SP/export" && cd "$SP/export" && unzip -o -q "C:/Users/LENOVO/Downloads/session-export-1791382480074.zip" && find . -type f -printf '%s\t%p\n' | sort -k2
```

```
1792131 ./e3dd8c12-86c1-477c-96a4-36c9acf65a74.jsonl
37      ./e3dd8c12-86c1-477c-96a4-36c9acf65a74/custom-title.json
824     ./local-session-state.json
94643   ./metadata.json
1792131 ./transcript.jsonl
```

Ran a command Inspect the structure of the exported transcript

```
SP="C:/Users/LENOVO/AppData/Local/Temp/claude/C--College-sem-v-se-lab4-57-match-three-gem/e3dd8c12-86c1-477c-96a4-36c9acf65a74/scratchpad" && cd "$SP/export" && python -B - <<'EOF'
import json, collections
rows = [json.loads(l) for l in open("transcript.jsonl", encoding="utf-8") if l.strip()]
print("records:", len(rows))
print("types:", collections.Counter(r.get("type") for r in rows))
keys = collections.Counter()
for r in rows: keys.update(r.keys())
print("keys:", dict(keys))
... (25 more lines not shown)
```

```
records: 429
types: Counter({'assistant': 142, 'user': 96, 'attachment': 30, 'custom-title': 30, 'agent-name': 28, 'atis-latch': 27, 'last-prompt': 25, 'queue-operation': 16, 'file-history-snapshot': 14, 'file-history-delta': 13, 'system': 5, 'mode': 3})
keys: {'type': 429, 'operation': 16, 'timestamp': 302, 'sessionId': 402, 'content': 8, 'parentUuid': 273, 'isSidechain': 273, 'promptId': 96, 'message': 238, 'uuid': 273, 'permissionMode': 6, 'origin': 6, 'promptSource': 6, 'turnOrigin': 6, 'turnPosition': 6, 'userType': 273, 'entrypoint': 273, 'cwd': 273, 'version': 273, 'gitBranch': 273, 'attachment': 30, 'rendered': 19, 'renderedRole': 19, 'las
blocks: {('user', 'str'): 6, ('assistant', 'thinking'): 40, ('assistant', 'tool_use'): 90, ('user', 'tool_result'): 90, ('assistant', 'text'): 12}
sidechain: Counter({False: 273, None: 156})
isMeta: Counter({None: 429})
```

```
---USER 2026-10-07T06:32:58.775Z meta= None | 'go through the pdf in this folder'
---USER 2026-10-07T06:49:50.650Z meta= None | "so I couldn't even run the game properly. it keeps crashing
saying c is not defined in board.py under the game folder. is this a bug they have given and want us to
solve?"
---USER 2026-10-07T06:53:47.896Z meta= None | "I didn't fork the repository and then clone it. I just
cloned the actual repository locally and I will committing to that. so after the fixes can we actually push
these commits and fixed working code to my other software engineering repository for submission?"
---USER 2026-10-07T12:42:20.661Z meta= None | 'i fixed the crash bug. I need to make a new commit for each
of the task given in the readme.md file. help me do it all'
---USER 2026-10-07T13:31:54.415Z meta= None | 'this is the path to the actual github repository I have
cloned locally: C:\\College\\sem-v\\se\\github\\SoftwareEngineering\\lab4.  this is lab4 (a separate folder
in that repo) where we have to push this changed code along with the videos and our chat as a pdf.'
---USER 2026-10-07T14:14:33.166Z meta= None | "<!-- reply 1 -->\n> Are those the videos you meant? If so,
there is nothing more to give me. If you have different ones, tell me their path or drop them in that same
folder, and I'll swap them in before pushing; the commit is local, so replacing them is
clean.\n\nyes\n\n\n"
```

Ran a command Print an ordered outline of the transcript records

```
SP="C:/Users/LENOVO/AppData/Local/Temp/claude/C--College-sem-v-se-lab4-57-match-three-gem/e3dd8c12-86c1-
477c-96a4-36c9acf65a74/scratchpad" && cd "$SP/export" && python -B - <<'EOF'
import json
rows = [json.loads(l) for l in open("transcript.jsonl", encoding="utf-8") if l.strip()]
for i, r in enumerate(rows):
    t = r.get("type")
    if t == "assistant":
        for b in r["message"]["content"]:
            if b["type"] == "text":
... (21 more lines not shown)
```

```
2 U.str 'go through the pdf in this folder'
3 attachment environment None system '[{\'content\': "<system-reminder>\\n# Environment\\nYou have been
invoked in the fol'
4 attachment model None system "[{'content': '<system-reminder>\\nYou are powered by the model named Opus
5.5. Th"
5 attachment deferred_tools_delta None system "[{'content': '<system-reminder>\\nThe following deferred
tools are now available "
6 attachment agent_listing_delta None system "[{'content': '<system-reminder>\\nAvailable agent types for
the Agent tool:\\n- cl"
7 attachment mcp_instructions_delta None system "[{'content': '<system-reminder>\\n# MCP Server
Instructions\\n\\nThe following MCP "
8 attachment skill_listing None system "[{'content': '<system-reminder>\\nThe following skills are
available for use with"
13 attachment session_context None user '[{\'content\': "<system-reminder>\\nAs you answer the user\'s
questions, you can use'
14 attachment date None system '[{\'content\': "<system-reminder>\\nToday\'s date is 2026-10-
07.\\n</system-reminder>'
15 attachment credential_org None None 'None'
16 attachment remote_session_change None user '[{\'content\': "<system-reminder>\\nAttribution for git
commits and pull requests y'
17 attachment prompt_snapshot None None 'None'
20 A.think 0
21 A.tool Read '{"file_path": "C:\\\\College\\\\sem-v\\\\se\\\\lab4\\\\57_match_three_gem\\\\Lab_4'
22 U.result str
23 A.tool Glob '{"pattern": "**/*"}'
24 U.result str
25 attachment prompt_snapshot None None 'None'
30 A.think 0
31 A.tool Read '{"file_path": "C:\\\\College\\\\sem-v\\\\se\\\\lab4\\\\57_match_three_gem\\\\Lab_4'
32 U.result ['text', 'document']
33 A.tool Bash '{"command": "python --version 2>&1; python -c \\"import importlib.util '
... (254 more lines not shown)
```

Ran a command Look at a few transcript details and check which Markdown libraries and browsers are available for PDF rendering

```
SP="C:/Users/LENOVO/AppData/Local/Temp/claude/C--College-sem-v-se-lab4-57-match-three-gem/e3dd8c12-86c1-
477c-96a4-36c9acf65a74/scratchpad" && cd "$SP/export" && python -B - <<'EOF'
import json, importlib.util as u
```

```
rows = [json.loads(l) for l in open("transcript.jsonl", encoding="utf-8") if l.strip()]
for i in (106, 141, 198):
    for b in rows[i]["message"]["content"]:
        print(i, b["type"], {k: (v if not isinstance(v, str) else v[:200]) for k, v in b.items() if k !=
"signature"})
r = rows[416]["message"]["content"][0]
print("ASK result:", repr(str(r.get("content"))[:300]))
... (5 more lines not shown)
```

```
106 thinking {'type': 'thinking', 'thinking': "Your crash fix is committed and pygame is installed, so I'll
now work through Tasks 1�4, testing and committing each separately. First, I'll untrack the
`__pycache__/*.pyc` files picked up by your com"}
141 thinking {'type': 'thinking', 'thinking': "Task 1 done. Next up, I'll implement task 2: escalating
chain-reaction multipliers (�1, �2, �3...) with a visible combo banner in the HUD.\n\n"}
198 thinking {'type': 'thinking', 'thinking': 'Task 2 is committed. Next up is task 3: implementing the
striped gem mechanic, where a 4+ match creates a gem that clears its row or column based on run direction,
with chain blasts into other special'}
ASK result: 'The user answered: "Everything is committed locally in SoftwareEngineering/lab4 (game code
with all task commits, plus both videos). How do you want to handle the push and the chat PDF?"="I will
give you the videos here. push it along with other things". Read the answers carefully \u2014 they may request'
toolUseResult keys (bash): ['stdout', 'stderr', 'interrupted', 'isImage', 'noOutputExpected']
image block: {'type': 'base64', 'data': '<32152 b64 chars>', 'media_type': 'image/png'}
libs: {'markdown': False, 'markdown_it': False, 'mistune': True, 'markdown2': False, 'fitz': True,
'pygments': True}
--- browsers ---
C:/Program Files (x86)/Microsoft/Edge/Application/msedge.exe
C:/Program Files/Google/Chrome/Application/chrome.exe
```

CLAUDE

Videos confirmed, so I'm on the last stretch: turning this chat's export into a readable PDF, then adding it to `lab4/` and pushing everything in one go. Building the PDF converter now.

Wrote a temporary helper script `scratch/chat_to_pdf.py` (226 lines, kept outside the repo)

Ran a command Convert the transcript to HTML, print it to PDF with headless Edge, and render sample pages for checking

```
SP="C:/Users/LENOVO/AppData/Local/Temp/claude/C--College-sem-v-se-lab4-57-match-three-gem/e3dd8c12-86c1-
477c-96a4-36c9acf65a74/scratchpad" && cd "$SP" && python -B -c "import mistune; print('mistune',
mistune.__version__)" && python -B chat_to_pdf.py export/transcript.jsonl chat.html && rm -f chat.pdf &&
MSYS_NO_PATHCONV=1 "C:/Program Files (x86)/Microsoft/Edge/Application/msedge.exe" --headless -
import fitz
doc = fitz.open("chat.pdf")
print("pages:", doc.page_count)
for n in (0, 1, doc.page_count // 2, doc.page_count - 1):
    doc[n].get_pixmap(dpi=70).save(f"page_{n + 1}.png")
    print("rendered page", n + 1)
EOF
```

```
mistune 3.3.4
prompts=6 actions=90 html=199 KB -> chat.html
[4552:24016:1007/194711.744:ERROR:chrome\browser\task_manager\providers\fallback_task_provider.cc:126]
Every renderer should have at least one task provided by a primary task provider. If a "Renderer" fallback
task is shown, it is a bug. If you have repro steps, please file a new bug and tag it as a dependency of
crbug.com/40528867.
517306 bytes written to file C:/Users/LENOVO/AppData/Local/Temp/claude/C--College-sem-v-se-lab4-57-match-
three-gem/e3dd8c12-86c1-477c-96a4-36c9acf65a74/scratchpad/chat.pdf
-rw-r--r-- 1 LENOVO 197609 517306 Oct  7 19:47 chat.pdf
warning: The `fitz` API is deprecated and will be removed in future. Use `import pymupdf` instead.
pages: 29
rendered page 1
rendered page 2
rendered page 15
rendered page 29
```

Checked a preview page of this PDF `scratch/page_1.png`

Checked a preview page of this PDF `scratch/page_15.png`

Adjusted a temporary helper script `scratch/chat_to_pdf.py`

Ran a command Regenerate the PDF with the diff spacing fixed and render a few key pages

```
SP="C:/Users/LENOVO/AppData/Local/Temp/claude/C--College-sem-v-se-lab4-57-match-three-gem/e3dd8c12-86c1-
477c-96a4-36c9acf65a74/scratchpad" && cd "$SP" && python -B chat_to_pdf.py export/transcript.jsonl
chat.html && rm -f chat.pdf && MSYS_NO_PATHCONV=1 "C:/Program Files
(x86)/Microsoft/Edge/Application/msedge.exe" --headless --disable-gpu --no-pdf-header-footer --print-to-
pdf="$SP/chat.pdf" "file:
import pymupdf
doc = pymupdf.open("chat.pdf")
print("pages:", doc.page_count)
hits = {}
for i, page in enumerate(doc):
    text = page.get_text()
    for key in ("Task 3: line-clear special gems", "COMBO", "Claude asked a question", "All four tasks are
done", "[email hidden]", "@gmail.com"):
... (8 more lines not shown)
```

```
prompts=6 actions=90 html=199 KB -> chat.html
pages: 24
{'COMBO': [8, 9, 17, 20], 'Task 3: line-clear special gems': [10], 'All four tasks are done': [19], 'Claude
asked a question': [23]}
rendered [19, 23, 10]
```

Checked a preview page of this PDF `scratch/check_19.png`

Checked a preview page of this PDF `scratch/check_23.png`

Checked a preview page of this PDF `scratch/check_10.png`

CLAUDE

The PDF renders well (24 pages: prompts, replies, code diffs, screenshots). Two small tweaks to the converter, then a fresh export so the PDF includes this last stretch too.

Ran a command Apply small wording and layout tweaks to the chat-to-PDF converter

```
SP="C:/Users/LENOVO/AppData/Local/Temp/claude/C--College-sem-v-se-lab4-57-match-three-gem/e3dd8c12-86c1-
477c-96a4-36c9acf65a74/scratchpad" && cd "$SP" && python -B - <<'EOF'
import pathlib
p = pathlib.Path("chat_to_pdf.py")
s = p.read_text(encoding="utf-8")
edits = [
    # Don't embed previews of this very PDF inside itself.
    ('        note = "screenshot rendered by the test script" if images else ""\n        return
action("Read", f"<code>{esc(path)}</code> {note}", imgs, failed)',
     '        if re.match(r"scratch/(page|check)_", path):\n            return action("Checked a preview
page of this PDF", f"<code>{esc(path)}</code>")\n        note = "screenshot rendered by the test script" if
images else ""\n        return action("Read", f"<code>{esc(path)}</code> {note}", imgs, failed)'),
... (11 more lines not shown)
```

Exported this chat to build this PDF

End of export. The chat was exported at this point to build this PDF; the only steps after it were adding this PDF to `lab4/` and pushing the lab repository to GitHub.